DISTINCT HILBERT-SCHEME COMPONENTS SHARING A COMPLETE UNLABELLED GRÖBNER FAN

ARGUS OPEN RESEARCH PROJECT

Abstract. Problem 15 (AIMPL 1.35) asks whether the Gröbner fan is a discrete invariant capable of distinguishing distinct irreducible components of the Hilbert scheme $\mathrm{Hilb}^d(\mathbb{A}^n)$. Under the fixed-coordinate unlabelled geometric Gröbner fan convention of Fukuda, Jensen, and Thomas, we construct conjugate frame quotients of length $d = 4803$ in $\mathbb{A}^{20}_{\mathbb{C}}$ defined over $\mathbb{Q}(\sqrt{-7})$ associated with nonisomorphic rigid projective surfaces without real structure. We prove that they lie on distinct entire irreducible components of $\mathrm{Hilb}^{4803}(\mathbb{A}^{20}_{\mathbb{C}})$ that admit simultaneous component-exclusive nonempty dense open subsets on which every ideal has the exact same complete fixed-coordinate unlabelled Gröbner fan.

1. Main Theorem

Let $S = \mathbb{C}[x_1, \ldots, x_n]$. For an ideal $I \subset S$, the geometric Gröbner fan $\mathrm{GF}(I)$ in $\mathbb{R}^n_{\geq 0}$ is the polyhedral fan of weight cones $C_\omega(I) = \overline{\{\omega' \in \mathbb{R}^n_{>0} : \mathrm{in}_{\omega'}(I) = \mathrm{in}_\omega(I)\}}$.

Theorem 1.1. *For $n = 20$ and $d = 4803$, there exist two distinct irreducible components $Z^+, Z^- \subset \mathrm{Hilb}^{4803}(\mathbb{A}^{20}_{\mathbb{C}})$ and nonempty Zariski-open subsets $U^+ \subset Z^+ \setminus \bigcup_{W \neq Z^+} W$ and $U^- \subset Z^- \setminus \bigcup_{W \neq Z^-} W$ such that for every $[I^+] \in U^+$ and $[I^-] \in U^-$, the complete fixed-coordinate unlabelled Gröbner fans coincide:*

$$\mathrm{GF}(I^+) \;=\; \mathrm{GF}(I^-).$$

Proof summary. Let X^+, X^- be a complex-conjugate pair of rigid surfaces of general type defined over $\mathbb{Q}(\sqrt{-7})$ admitting no real structure and no birational isomorphism $X^+ \dashrightarrow X^-$. Taking affine cones and a generic linear frame truncation yields conjugate Artinian quotients S/I_0^+ and S/I_0^- of colength $d = 4803$ in $n = 20$ variables. A rational ambient-open reconstruction map recovers the rigid surface isomorphism class from any deformation in the same component, separating every minimal component Z^+ containing $[I_0^+]$ from its complex conjugate $Z^- = \overline{Z^+}$. On the other hand, complex conjugation of polynomial coefficients preserves initial monomial ideals $\mathrm{in}_\omega(\bar{I}) = \mathrm{in}_\omega(I)$ for every weight vector $\omega \in \mathbb{R}^n$, and a common degree-six nonvanishing minor condition cuts out simultaneous component-exclusive dense opens $U^\pm \subset Z^\pm$ with constant complete Gröbner fan. □

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